End-to-End Optimized Lossless Hyperspectral Image Compression With Bit Partition

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***Abstract*—Lossless compression of hyperspectral images (HSI) is crucial in applications such as scientific archiving, geological exploration, and material identification, where spectral fidelity must be strictly preserved. However, the extremely high-bit-depth and strong spectral redundancy of HSI make existing learning-based image compression methods mainly designed for RGB images ineffective, as they fail to model high-dimensional dependencies and suffer from inefficiency under high-bit-depth sampling. To address these challenges, we propose LHIC-BP, the first end-to-end optimized lossless compression framework specifically tailored for hyperspectral data. The core idea is to optimize bitrate directly through conditional probability modeling, using a discrete logistic mixture model to estimate pixel-level probability distributions. To efficiently handle high-bit-depth hyperspectral inputs, LHIC-BP introduces a bit partition strategy that decomposes the image into the most significant plane (MSP) and least significant plane (LSP). The MSP captures dominant intensity structures with low complexity, while the LSP retains fine-grained residual details that require stronger contextual modeling. To further enhance compression efficiency, we design a dedicated entropy model that leverages both spatial and spectral priors to refine conditional probability estimation for MSP and LSP. In the spatial dimension, a parallel spatial condition aggregation module (PSCAM) is introduced to efficiently eliminate local redundancy. Meanwhile, in the spectral dimension, an RWKV-guided spectral condition extraction module (RSCEM) is used to capture long-range interband dependencies with linear complexity. Extensive experiments on the HyspecNet-11K datasets show that LHIC-BP consistently surpasses state-of-the-art methods, achieving superior compression ratios and validating its effectiveness for hyperspectral lossless compression.**

***Index Terms*—Conditional coding, hyperspectral image (HSI), learning-based compression, lossless compression, remote sensing.**

I. Introduction

HYPERSPECTRAL imaging (HSI) has become an indispensable tool in modern Earth observation owing to its ability to capture fine spectral details across hundreds of contiguous bands. This unique capability supports a wide range of critical applications, including high-precision urban monitoring, environmental protection, and climate change assessment [1], [2], [3]. With the rapid deployment of hyperspectral satellite constellations, advances in spatial, spectral, and radiometric resolutions have led to the generation of unprecedented volumes of high-dimensional data. Consequently, the development of efficient compression algorithms has emerged as a pressing need to overcome storage constraints, reduce transmission bandwidth, and ensure timely data accessibility.

Conventional image compression techniques were originally designed for natural three-channel RGB images, with the primary goal of reducing storage and transmission costs by removing redundancies. Depending on whether information loss is permitted, these techniques are typically divided into lossy and lossless schemes. Lossy methods achieve substantial bitrate reductions by discarding perceptually less significant details, with representative standards, including JPEG [4], WEBP [5], and BPG [6]. With the rise of deep learning, end-to-end lossy compression methods [7], [8], [9] have demonstrated remarkable improvements in both visual quality and compression efficiency. In contrast, lossless compression completely preserves the original information and is indispensable in high-fidelity applications such as medical imaging and scientific remote sensing. Over the past decades, a variety of lossless standards have been developed, including PNG, JPEG 2000 [10], and JPEG XL [11]. More recently, deep learning-based approaches [12], [13], [14] have emerged, showing strong potential to further advance this field and opening up new research opportunities.

Compared with RGB image compression, hyperspectral image compression faces more severe challenges, primarily due to the extremely high spectral resolution that introduces substantial dimensional redundancy and the high-bit-depth sampling that increases coding complexity. Consequently, hyperspectral image compression has been explored far less extensively than RGB image compression. Similar to other image compression tasks, hyperspectral image compression can also be divided into lossy and lossless schemes, with most existing studies focusing on lossy approaches [15], [16], [17], [18], [19]. However, lossy methods inevitably introduce information loss, which may negatively affect downstream tasks such as classification, object detection, and change detection.

In contrast, lossless compression guarantees complete fidelity and reversibility, making it particularly suitable for applications that demand strict data integrity, such as agricultural monitoring, mineral exploration, and environmental assessment. The main challenge in hyperspectral lossless compression lies in effectively eliminating both spatial and spectral redundancies. Existing methods are largely based on handcrafted designs, such as predictive or transform coding,

Received 25 November 2025; revised 3 February 2026; accepted 10 February 2026. Date of publication 24 February 2026; date of current version 5 March 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62572357. *(Corresponding author: Zhenzhong Chen.)*

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Digital Object Identifier 10.1109/TGRS.2026.3667533

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which struggle to fully exploit high-dimensional correlations and therefore achieve only limited performance [20], [21], [22].

Deep learning-based lossless compression for RGB images, by comparison, has demonstrated strong modeling capacity and the advantages of end-to-end frameworks. However, due to the multiband and high-dimensional characteristics of hyperspectral data, these techniques cannot be directly applied, and research on deep learning-based hyperspectral lossless compression is still in its infancy. For example, LineRWKV [23] introduces a predictive coding framework in which a neural network predicts the current pixel value and encodes only the residuals, achieving modest improvements. Nevertheless, such methods still fail to fully exploit the contextual modeling capability of deep neural networks, primarily because they lack end-to-end optimization toward the final compression objective, which fundamentally limits their achievable performance.

Motivated by the aforementioned observations, we propose a novel framework termed lossless hyperspectral image compression with bit partition (LHIC-BP). The framework follows an end-to-end deep learning paradigm and employs a joint optimization strategy of conditional probability modeling and entropy coding to achieve lossless compression. Specifically, we adopt a discrete logistic mixture model to characterize the probability distribution of each pixel and generate the bitstream via entropy coding. Considering that HSI typically possess high radiometric resolution, directly computing the cumulative distribution function (cdf) over the full range incurs prohibitive memory and computational costs. To address this issue, we introduce a bit partition strategy: the original high-bit-depth image is decomposed into most significant planes (MSPs) and least significant planes (LSPs). These two components are modeled separately, and the MSP prior is incorporated into LSP probability prediction to mitigate the randomness of low-order bits.

For conditional feature extraction, we design a dedicated entropy model tailored to the 3-D structure of hyperspectral image cubes. In the spatial dimension, we introduce a parallel spatial condition aggregation module (PSCAM), which integrates masked convolution-based conditional aggregation with a block-wise diagonal autoregressive scheme (BDAS) to eliminate local spatial redundancy and construct autoregressive contexts in a near-parallel manner while preserving causality. In the spectral dimension, as illustrated in Fig. 1, correlation analysis reveals pronounced long-range interband dependencies that cannot be effectively captured by conventional convolution-based models. To efficiently exploit this spectral redundancy, we develop an RWKV-guided spectral condition extraction module (RSCEM), which leverages recurrent-style weighted key–value (WKV) operations to model long-range spectral context with linear computational and memory complexity. Finally, the estimated probability distributions for both MSP and LSP are fed into the entropy coder to generate the final lossless bitstream.

In summary, our contributions are summarized as follows.

1) We present LHIC-BP, the first end-to-end optimized framework tailored for hyperspectral lossless compression, jointly optimizing conditional probability modeling and entropy coding to directly minimize bitrate.
2) We propose a bit partition strategy that decomposes hyperspectral data into MSP and LSP, enabling efficient modeling of high-bit-depth data with reduced computational cost for cdf while preserving full fidelity.
3) We design PSCAM to remove local spatial redundancy while constructing autoregressive contexts in a near-parallel manner. The module internally employs a BDAS to preserve causality and significantly improve compression efficiency in the spatial dimension.
4) We develop RSCEM to capture long-range spectral correlations efficiently, exploiting dependencies across spectral bands with linear complexity. Moreover, we build a streaming coding scheme based on the recursive formula of RWKV.

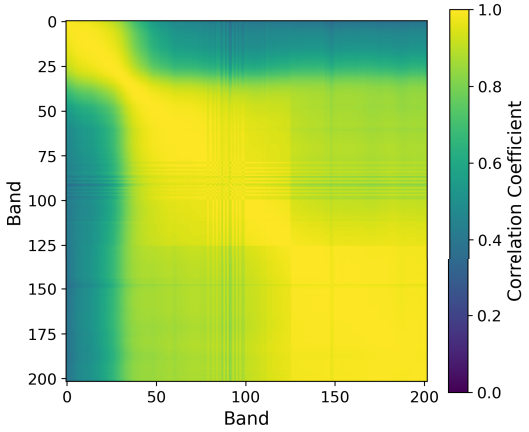


Fig. 1. Visualization of band correlation matrix, which shows that long-distance dependencies exist between different bands.

The rest of this article is organized as follows. Section II reviews the related work on hyperspectral image compression. Section III presents the proposed LHIC-BP framework in detail. Section IV describes the experimental setup, reports the results with corresponding analyses, and discusses the impact of each component in the proposed method. Finally, Section V concludes this article and outlines potential future research directions.

II. Related Work

In this section, we provide a review of image compression research across the following three subdomains: end-to-end image compression, lossy hyperspectral image compression, and lossless hyperspectral image compression.

A. End-to-End Image Compression

Learned image compression methods were initially inspired by the variational autoencoder (VAE) [24] framework, which learns latent representations and entropy models in an end-to-end encoder–decoder optimized under a rate–distortion (R–D) objective. Ballé et al. [7] proposed a fully factorized entropy model to support differentiable quantization and approximate entropy-coding optimization. Minnen et al. [8]

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incorporated hyperprior models to improve entropy modeling and reduce cross entropy. Subsequent works further enhanced entropy models via context modeling. Minnen and Singh [9] adopted channel-wise autoregressive modeling, Cheng et al. [25] used self-attention for global dependency modeling, and He et al. [26] proposed a checkerboard autoregressive approach to accelerate coding and decoding. Many works have also explored improvements in the transform modules within the VAE architecture [27], [28].

In comparison, the present work is more closely related to the field of learned lossless image compression, where, in theory, any probabilistic model can be combined with an entropy coder to compress data losslessly into compact bitstreams. According to Shannon's information theory [29], the lower bound of the bit rate is determined by the accuracy of the probability model—that is, the more precise the probability estimation, the lower the achievable bit rate. Depending on how probability models are estimated, representative learned lossless compression methods can be categorized into three main classes.

Autoregressive Models: These methods decompose the joint probability distribution into conditional distributions. Oord et al. [12] first proposed PixelCNN, which employs masked convolutions for conditional probability modeling. Ryder et al. [30] introduced a subimage regression approach, while Rhee et al. [31] developed a method that decomposes images into high- and low-frequency components followed by autoregressive modeling.

Flow-Based Models: Ho et al. [32] proposed the local bits-back algorithm, which leverages flow models to model latent structures of images. Hoogeboom et al. [33] introduced integer discrete flows for reversible transformations of raw images, followed by step-wise entropy coding of the transformed data. Berg et al. [34] demonstrated the feasibility of modeling arbitrary distributions using integer discrete flows, and Zhang et al. [35] proposed modular scale transforms for enhanced flexibility.

VAE-Based Models: Mentzer et al. [13] proposed L3C, the first practical full-resolution learned lossless compression method, which employs a parallelizable hierarchical probability model. Townsend et al. [36] presented a bits-back coding scheme to losslessly encode both latent and original image information while enabling end-to-end optimization via the variational lower bound. Kingma et al. [37] introduced Bit-Swap, which reorganizes latent representations and optimizes the bits-back process for improved speed and compression.

However, all these methods are primarily designed for three-channel RGB images and model interband relationships using a simple linear formulation [14], [38]. In contrast, HSI exhibit a far more complex structure, particularly due to the substantial redundancies along the spectral dimension and high bit-depth, which renders existing approaches directly inapplicable to hyperspectral lossless compression.

B. Lossy Hyperspectral Image Compression

Hyperspectral image (HSI) compression methods have traditionally relied on transform coding, where pixel-domain HSI data are converted into a latent representation (e.g., frequency domain) for subsequent processing. Commonly used linear transforms include the discrete cosine transform (DCT) and the wavelet transform, both of which are based on linear and orthogonal basis functions. In recent years, learning-based approaches have shown remarkable promise for HSI compression. For instance, Dua et al. [15] and La Grassa et al. [16] were among the first to employ autoencoder (AE) architectures for lossy HSI compression. Building upon the VAE framework, Guo et al. [39] incorporated a hyperprior model [8] and replaced the conventional Gaussian prior with a Student's t-distribution to better model the latent variables. More recently, Guo et al. [18] proposed a contrastive learning-enhanced framework to preserve spectral fidelity during compression, Kuester et al. [19] introduced a two-stage AE designed to adapt to data from heterogeneous sensors. Although these lossy compression methods achieve impressive performance, their inherent information loss causes irreversible degradation, which can negatively impact downstream applications. This limitation motivates the exploration of lossless compression methods for HSI, where preserving complete information is critical.

C. Lossless Hyperspectral Image Compression

Lossless hyperspectral image compression has long been a central topic in image compression, aiming to remove both spatial and spectral redundancies. Existing methods are mainly grouped into predictive coding and transform coding. Predictive coding is a widely used paradigm that predicts each pixel from its neighbors and encodes the residual to exploit spatial correlations. To improve prediction accuracy, Mielikainen and Huang [40] applied K-means clustering to group spectral lines and then estimated prediction parameters via least-squares optimization. Tate [41] further enhanced predictive performance through band reordering. More recently, the CCSDS123.0-B-2 standard [21] adopted adaptive predictive coding, dynamically selecting prediction directions across spatial and spectral dimensions while maintaining relatively low complexity. Transform coding provides another major class of methods, projecting HSI data into low-dimensional spaces to compact energy and reduce interspectral redundancy. For instance, JPEG2000 [10] is effective for spatial redundancy, while spectral redundancy is often addressed by reversible transforms such as the Karhunen–Loève transform (KLT) [22] and integer wavelet transforms [20]. Despite their success, these approaches heavily rely on manually designed predictors/transforms and heuristic modeling rules, which may limit their flexibility in capturing complex correlations in high-dimensional data.

In contrast, learning-based lossless HSI compression remains relatively underexplored. To the best of our knowledge, the only representative work is LineRWKV [23], which combines predictive coding with a neural network predictor. Although it demonstrates the promise of deep learning, it uses the network only for prediction and encodes residuals with a conventional codec, without end-to-end rate-oriented optimization, thus limiting the achievable gains.

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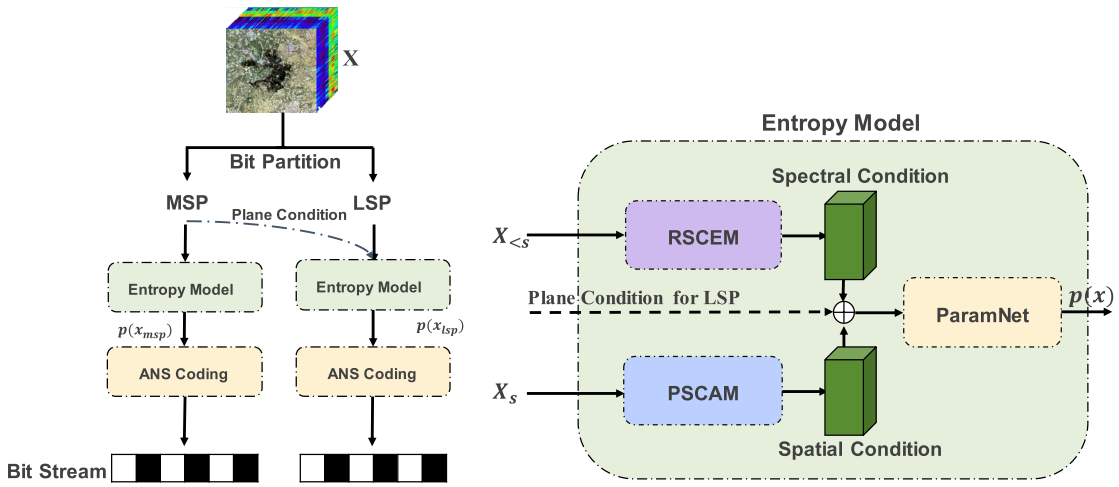


Fig. 2. Overview of the proposed LHIC-BP framework, which consists of bit partition, entropy model, and entropy coding. The entropy model has two components, in which PSCAM is used to capture spatial conditions and RSCEM is used to extract spectral conditions. For LSP coding, MSP is used as a plane condition to enhance coding performance.

III. Proposed Method

A. Overview

In this section, we provide an overview of the proposed LHIC-BP framework. As shown in Fig. 2, given an S-band hyperspectral image $X = (X_1, X_2, \ldots, X_S)$, we first perform bit partitioning to obtain two subimages, MSP $X^M = (X_1^M, \ldots, X_S^M)$ and LSP $X^L = (X_1^L, \ldots, X_S^L)$, thereby alleviating the excessive memory consumption incurred by subsequent cdf calculations. Under a unified "conditional probability modeling–entropy coding" paradigm, separate probabilistic models are trained end-to-end for MSP and LSP. For each spectral band, PSCAM extracts spatial conditions from the already encoded neighborhood within the same band, while RSCEM captures spectral dependencies from previously encoded bands; moreover, the reconstructed MSP is used to construct a bit-plane condition for the LSP branch. Given these conditions, ParamNet predicts the parameters of the discrete probability distribution at each location, and MSP/LSP symbols are entropy-coded to form the bitstream.

Decoding strictly follows the same order and conditioning as encoding, adopting a two-stage pipeline of "MSP first, then LSP." The decoder first reconstructs MSP progressively in a predefined diagonal spatial order and spectral band order. At each step, it computes the symbol distribution based on the decoded spatial neighborhood and the cross-band states maintained by RSCEM; the MSP value is then decoded and cached as context. After MSP is fully recovered, the corresponding plane condition is constructed and provided to the LSP branch, where LSP is decoded using the same spatial and spectral order. Finally, MSP and LSP are merged bitwise to produce the reconstructed lossless image $\hat{X}$.

1) Conditional Coding With Logistic Mixture Model: Motivated by the success of modern neural compression techniques, we adopt the paradigm of conditional coding to enhance modeling efficiency. Specifically, instead of directly predicting the value of the next pixel, the network aims to model its conditional probability distribution given the conditional information C. In contrast to traditional predictive coding, which assumes a Dirac δ distribution centered at the predicted value and thus ignores the true statistical variability of the pixel, conditional coding directly models the full distribution $p(X|C)$. By leveraging the correlation between C and the target pixel, the model accurately captures the pixel's spatial–spectral dependencies and provides a more precise probability estimate for entropy coding, which leads to improved compression performance and reduced coding redundancy.

Moreover, due to the high-bit-depth characteristics of HSI, directly applying a softmax activation function at the output to represent a dense probability distribution suffers from the softmax bottleneck [42]. To address this issue, our framework follows the approach in [38] and integrates a logistic mixture model. Specifically, the network learns the parameters of k logistic distributions, including means $\{\mu_1, \mu_2, \ldots, \mu_k\}$, scales $\{s_1, s_2, \ldots, s_k\}$, and mixture weights $\{\pi_1, \pi_2, \ldots, \pi_k\}$, and the probability distribution of the target pixel x is given by the following:

$$p_\theta(x) = \sum_{i=1}^{K} \pi_i \left[\sigma\left(\frac{x + 0.5 - \mu_i}{s_i} \right) - \sigma\left(\frac{x - 0.5 - \mu_i}{s_i} \right) \right] \quad (1)$$

where σ indicates the logistic sigmoid function. With (1), the cross entropy between the learned distribution p_θ and the real unknown distribution p can be formulated as follows:

$$H(p, p_\theta) = \mathbb{E}_{X \sim p} \left[\sum_{s=1}^{S} -\log p_\theta(X_s | C_s) \right] \quad (2)$$

which indicates the corresponding compression performance of X. Particularly, for a given sth band X_s, condition C_s mainly includes previous spectral condition C_s^{spe} and local spatial condition C_s^{spa}. Thus, 2 can be rewritten as follows:

$$H(p, p_\theta) = \mathbb{E}_{X \sim p} \left[\sum_{s=1}^{S} -\log p_\theta\left(X_s \left| C_s^{spe}, C_s^{spa} \right. \right) \right]. \quad (3)$$

The parameter generation network (ParamNet), as illustrated in Fig. 3, primarily consists of 2-D convolutions with kernel

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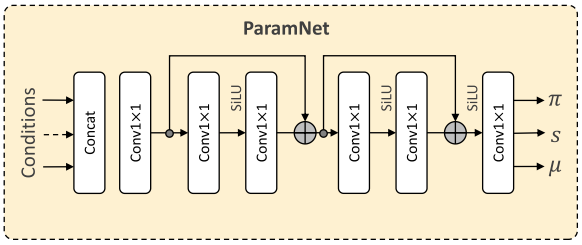


Fig. 3. Structure of ParamNet. The parameter of logistic mixture model π, μ, and s is generated by ParamNet.

size of 1 and SiLU activation functions. The use of 1×1 convolution layers allows the modeling of channel correlations while maintaining causality.

2) Bit Partition: Building on the conditional coding framework with a logistic mixture model, we encounter substantial computational challenges when processing 16-bit HSI, since the wide quantization range (0–65 535) makes per-pixel probability estimation expensive. Specifically, entropy coding requires computing the cdf at each pixel. A naive implementation would construct an integer array of length 65 537 per pixel; to avoid arithmetic overflow and preserve coding accuracy, this array must be stored in 32-bit integers. The resulting memory footprint is prohibitive. Even for a modest 32×32 image, the cdf arrays would require approximately 36 GB of memory, rendering the straightforward approach impractical. To address this issue, we introduce a bit-partition strategy that decomposes a high-bit-depth image into two lower bit-depth subimages, thereby effectively circumventing the memory bottleneck.

For the input hyperspectral image X, the bit partition operation is defined as follows:

$$X^M = \lfloor X/2^d \rfloor, \quad X^L = X \bmod 2^d \tag{4}$$

where X^M represents the MSP, X^L corresponds to the LSP, and d is a hyperparameter that defines the bit-depth split position. However, bit partition introduces a critical issue. LSP exhibits higher data randomness, which could degrade performance if directly encoded using conditional probability models. To address this, inspired by [14], LHIC-BP first decodes the MSP and then utilizes it as a bit-plane condition to guide the context modeling of the LSP. We adopt a U-Net structure, similar to [14], for bit-plane condition modeling and producing the plane condition C_s^{plane} for each sth band X_s^L in the LSP. With the proposed bit partition strategy, the cross entropy between the learned distribution p_θ and the real unknown distribution p can be rewritten as follows:

$$\begin{aligned} H(p, p_\theta) = &\mathbb{E}_{X\sim p}\left[\sum_{s=1}^{S} -\log p_\theta\left(X_s^L | C_s^{L,spe}, C_s^{L,spa}, C_s^{plane}\right)\right] \\ &+ \mathbb{E}_{X\sim p}\left[\sum_{s=1}^{S} -\log p_\theta\left(X_s^M | C_s^{M,spe}, C_s^{M,spa}\right)\right]. \end{aligned} \tag{5}$$

We adopt (5) as the loss function for the entire network, enabling end-to-end optimization by minimizing its value.

Now that the end-to-end architecture of LHIC-BP has been defined, the remaining problem is how to efficiently obtain the conditional priors C_s for effective probability modeling. Section III-B describes the aggregation of spatial condition C_s^{spa}, while Section III-C presents the extraction of spectral condition C_s^{spe}.

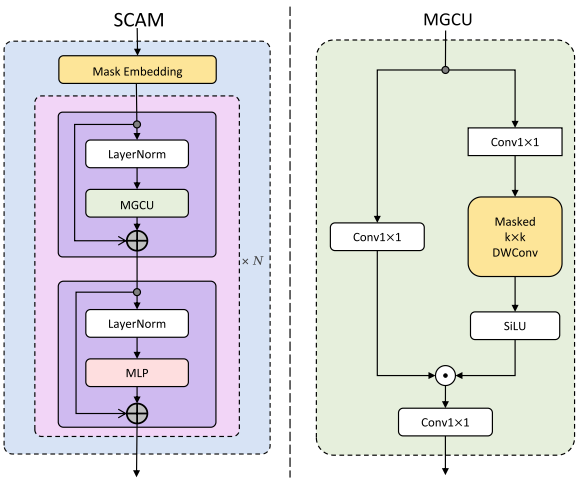


Fig. 4. Architecture of SCAM and its internal MGCU block. SCAM aggregates spatial context under causality constraints using stacked MGCUs, each combining masked depthwise convolution with gated feature mixing.

B. Parallel Spatial Condition Aggregation Module

1) Spatial Condition Aggregation Module: Given the relatively low ground spatial distance of HSI, our primary focus is on exploiting local spatial redundancies. First, we adopt a spatial condition aggregation module (SCAM) for spatial autoregressive modeling, inspired by [43]. As illustrated in Fig. 4, the original (C, H, W) hyperspectral image is treated as C 2-D images of size (H, W), which are fed into the network to extract corresponding feature representations. Within SCAM, a masked convolution is first applied to the input for causal embedding. Specifically, a 3×3 masked convolution is employed to aggregate local information. This is followed by N masked gated context units (MGCUs), which further fuse the local spatial features. Following the design principles of MetaFormer [44], SCAM utilizes MGCUs as the Spatial Mixer for spatial information fusion and subsequently applies an MLP to integrate channel-wise information. The MLP consists of two 1×1 convolutions with the SiLU activation function. All normalization layers are implemented as LayerNorm to prevent potential information leakage.

The core component of SCAM is MGCU. To address the diverse characteristics of different bands while considering the inherently low ground spatial distance of HSI, MGCU focuses on local feature modeling instead of global dependencies. Specifically, a depthwise masked convolution with kernel size $k \times k$ is employed to extract local contextual information, followed by a gating mechanism that adaptively modulates the feature responses. The operation of MGCU can be formulated as follows:

$$\begin{aligned} A(X) &= \mathrm{DWConv}_{k\times k}(W_A X, M) \\ \mathrm{MGCU}(X) &= \sigma(A(X)) \odot (W_V X) \end{aligned} \tag{6}$$

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where W_A and W_V denote linear mappings implemented by 1×1 convolutions, $\sigma(\cdot)$ represents the SiLU activation function, $\odot$ denotes the Hadamard product, and M indicates the causal mask that restricts the receptive field to the available decoding context.

Through this formulation, the depthwise masked convolution captures fine-grained local structures, the nonlinear activation converts them into adaptive gating weights, and the Hadamard product modulates projected features in a content-aware manner. Overall, MGCU effectively integrates local context aggregation, adaptive gating, and feature modulation, enabling the network to model spatially constrained dependencies while maintaining spectral fidelity, as all operations are strictly intraband and avoid any cross-band feature mixing, thereby preventing spectral crosstalk.

2) Blockwise Diagonal Autoregressive Scheme: In the SCAM, masked convolutions are employed to enforce causal modeling. However, this design inherently requires pixel-wise sequential processing during encoding and decoding, which severely impacts practical efficiency. For an image of size $H \times W$, a total of $H \cdot W$ autoregressive steps are needed to generate the final result.

To address this issue, we further introduce a parallel acceleration strategy termed BDAS, which combines local autoregression [45] [46] with diagonal masking to significantly reduce computational overhead. Specifically, given a full-resolution image of size $H \times W$, we first partition it into nonoverlapping blocks of size P. Autoregressive modeling is then performed in parallel across blocks, reducing the number of sequential steps from $H \cdot W$ to P^2.

Furthermore, to minimize the remaining intrablock computation, we adopt a diagonal autoregressive ordering. We redesign the masks to follow a diagonal pattern, enabling dependencies to propagate more efficiently within each block. As illustrated in Fig. 5, while the original raster-order masked convolutions require 25 sequential steps to model the dependencies shown in Fig. 5(a), the proposed BDAS reduces this to only seven steps as shown in Fig. 5(b). For example, for a 128×128 image with block size $P = 32$, the total number of autoregressive steps is reduced from 16 384 to only 63. This dramatically improves the practical speed of both encoding and decoding without sacrificing causality.

By integrating BDAS with SCAM, we obtain the parallel spatial conditional aggregation module (PSCAM), combining local autoregression and parallelization to achieve efficient spatial contextual modeling without compromising causal context.

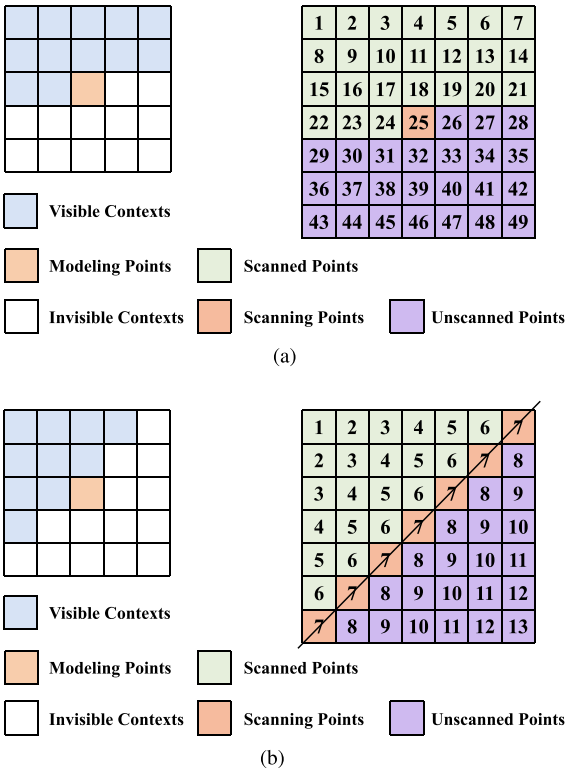


Fig. 5. Two modes for spatial autoregression, where 5(a) indicates raster scanning order and 5(b) indicates the proposed diagonal scanning order. (a) Raster. (b) Diagonal.

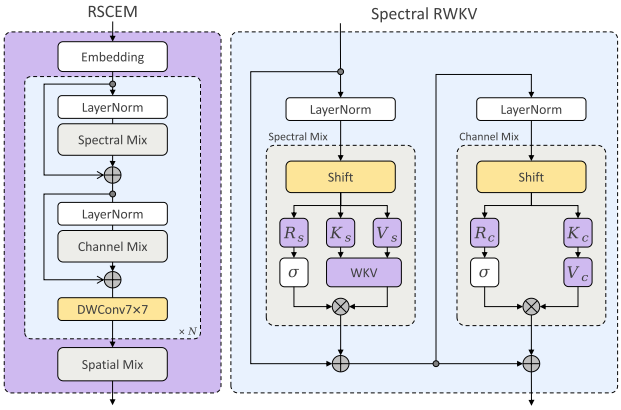


Fig. 6. Architecture of the RSCEM, integrating spectral, channel, and spatial mix blocks. The right panel expands the RWKV-based mixers, where shift and WKV operations capture long-range dependencies with linear complexity.

C. RWKV-Guided Spectral Condition Extraction Module

Due to the long-range dependencies present across spectral bands in HSI, we introduce the RWKV mechanism for spectral condition extraction, inspired by LineRWKV [23]. Unlike conventional attention mechanisms, whose computational complexity grows quadratically with sequence length, RWKV achieves linear scaling, making it particularly suitable for handling long sequences such as those in hyperspectral data. As illustrated in Fig. 6, the overall workflow begins with an input tensor, which is first transformed via a standard 2-D convolutional layer into an embedded feature representation. This representation is then processed through N SpectralRWKV blocks to perform contextual modeling across spectral dimensions.

1) Parallel Spectral Condition Extraction: The original RWKV supports two operational forms: a parallel form, optimized for efficient training, and a recursive form, designed for inference. Here, we first introduce the parallel form, which enables parallel spectral condition extraction during training. As shown in Fig. 6, RSCEM is composed of three key components: spectral mixing, channel mixing, and spatial mixing.

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a) Spectral Mixing: The spectral mixing module is designed based on the time mixing mechanism in the original RWKV architecture and is used to model long-range dependencies along the spectral dimension. Given an input feature tensor $F_x \in \mathbb{R}^{S\times H\times W\times F}$, where F denotes the feature dimension and S represents the number of spectral bands, we first reshape it to $F_x \in \mathbb{R}^{HW\times S\times F}$ to facilitate spectral sequence modeling at each spatial location. Subsequently, LayerNorm and a shift operation are applied to the input features. The specific operations are formulated as follows:

$$\begin{aligned} r_t &= W_r \cdot (\mu_r \odot x_t + (1-\mu_r)\odot x_{t-1}) \\ k_t &= W_k \cdot (\mu_k \odot x_t + (1-\mu_k)\odot x_{t-1}) \\ v_t &= W_v \cdot (\mu_v \odot x_t + (1-\mu_v)\odot x_{t-1}) \end{aligned} \tag{7}$$

where W_r, W_k, and W_v are all learnable parameters and μ_r, μ_k, and μ_v are all hyperparameters. Here, r_t, k_t, and v_t are the receptance, key, and value produced by the spectral mixing module at the tth spectral band, respectively; r_t is used for gating, while (k_t, v_t) are used for spectral-dependency extraction. Then, the attention values are computed through a WKV operation. We adopt the parallelized formulation of WKV proposed in [47] during the training phase, to enable efficient modeling in a fully parallel manner

$$wkv_t = \frac{\sum_{i=1}^{t-1} e^{(-(t-1-i)w+k_i)}\odot v_i + e^{(u+k_t)}\odot v_t}{\sum_{i=1}^{t-1} e^{(-(t-1-i)w+k_i)} + e^{(u+k_t)}} \tag{8}$$

where w and u are learnable parameters. After the WKV operation, the spectral mixing module applies gating to produce the output. Specifically, a sigmoid is applied to the shifted tensor R to generate gates, which are then multiplied element-wise with the WKV output. Finally, a linear projection produces the final result. The process is summarized as follows:

$$o_t = W_o \cdot (\sigma(r_t)\odot wkv_t). \tag{9}$$

b) Channel Mixing: The channel mixing module follows the spectral mixing module and is designed to further refine the modeling of dependencies. Similar to the spectral module, it begins with a LayerNorm operation, followed by a shift operation. Finally, it performs a gating operation. The entire process can be formulated as follows:

$$\begin{aligned} r'_t &= W'_r \cdot \left(\mu'_r \odot x_t + \left(1-\mu'_r\right)\odot x_{t-1}\right) \\ k'_t &= W'_k \cdot \left(\mu'_k \odot x_t + \left(1-\mu'_k\right)\odot x_{t-1}\right) \\ o'_t &= \sigma\left(r'_t\right)\odot\left(W'_v \cdot \max\left(k'_t, 0\right)^2\right) \end{aligned} \tag{10}$$

where W'_r, W'_k, and W'_v are also learnable parameters, μ'_r and μ'_k are some hyperparameters, and σ is the sigmoid function. In essence, the channel mixing module plays a role similar to the position-wise FFN in transformers; it applies a nonlinear channel transformation to each token and uses the gating term $\sigma(r'_t)$ to modulate the transformed features.

c) Spatial Mixing: The above two components are adaptations of the original RWKV architecture tailored to the hyperspectral image structure and together constitute the adopted SpectralRWKV module. However, using only these two modules constrains the context of the current pixel to the preceding spectral bands at a limited set of spatial positions, which may result in insufficient contextual information. To mitigate this limitation, a spatial mixing module is introduced after the spectral mixing and channel mixing modules to further broaden the available context. In practice, a lightweight U-Net style module used in [14] is employed to expand the receptive scope of spectral condition extraction by fusing features from previously processed bands across different spatial locations. For the output of the SpectralRWKV module, this module aggregates its spatial neighborhood and produces per-band spectral conditions. As illustrated in Fig. 7, the adopted U-Net comprises 3×3 convolution, ResBlock, and attention modules and uses skip connections to fuse encoder and decoder features. This design effectively expands the spectral-condition context from a confined local region to nearly all previously processed pixels, thereby enlarging the receptive field and enhancing the spectral contextual modeling capability of the proposed entropy model.

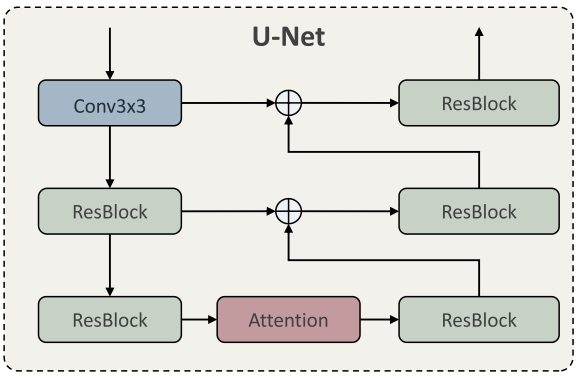


Fig. 7. Architecture of the U-Net used in spatial mixing.

2) Recursive Inference Form: During inference, RSCEM employs a recursive computation scheme, analogous to recurrent neural networks (RNNs), to significantly reduce the memory footprint of the WKV operation. Specifically, the process of WKV can be described in a recursive form, and the process of spectral mixing can be rewritten as follows:

$$\begin{aligned} a_0, b_0 &= 0 \\ r_t &= W_r \cdot (\mu_r \odot x_t + (1-\mu_r)\odot x_{t-1}) \\ k_t &= W_k \cdot (\mu_k \odot x_t + (1-\mu_k)\odot x_{t-1}) \\ v_t &= W_v \cdot (\mu_v \odot x_t + (1-\mu_v)\odot x_{t-1}) \\ wkv_t &= \frac{a_{t-1} + e^{u+k_t}\odot v_t}{b_{t-1} + e^{u+k_t}} \\ a_t &= e^{-w}\odot a_{t-1} + e^{k_t}\odot v_t \\ b_t &= e^{-w}\odot b_{t-1} + e^{k_t} \\ o_t &= W_o \cdot (\sigma(r_t)\odot wkv_t). \end{aligned} \tag{11}$$

All parameters are obtained during parallel training, and no further modification to the computational flow is required. As a result, the entire RWKV mechanism can be equivalently reformulated as an RNN-style recursive process. From its recurrence, it is evident that RWKV only needs to retain a_t and b_t, the previous hidden feature, and the newly computed feature at each step. This leads to substantially lower memory

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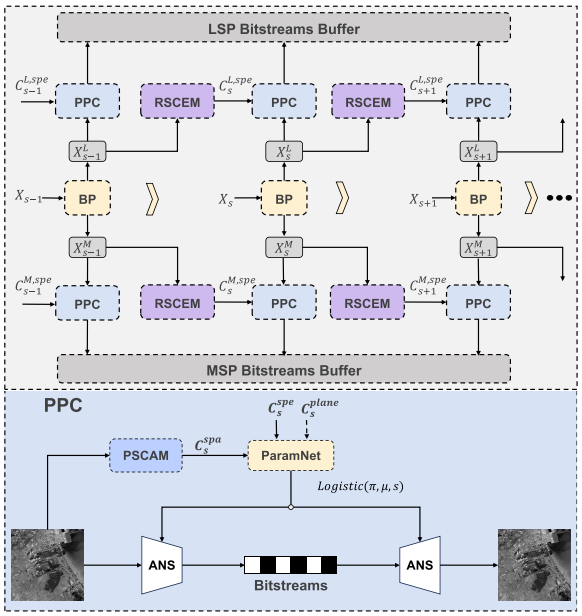


Fig. 8. Streaming coding scheme during inference mode for the proposed LHIC-BP, where PPC denotes the combination of PSCAM, ParamNet, and ANS coding.

usage than attention-based mechanisms, making inference highly efficient.

Building upon this, our LHIC-BP framework adopts a streaming coding paradigm during inference, as illustrated in Fig. 8. Specifically, the first spectral band is encoded without any spectral prior; hence, both the MSP and LSP are encoded solely under spatial and plane-based conditions. For subsequent spectral bands, the spectral conditions are computed recursively via RSCEM using the above formulation with pretrained parameters. These spectral conditions are then fused with the spatial condition generated by the PSCAM and the plane condition predicted by the U-Net. Then, ParamNet is used to produce the distribution parameters. Finally, the encoded data are converted into bit-streams using asymmetric numeral systems (ANSs) [48].

The decoding process mirrors the encoding procedure in reverse order. Since LSP reconstruction depends on the plane condition constructed from MSP, MSP is decoded first, followed by LSP. Specifically, within each band, symbols are decoded progressively in a diagonal scan order. At each band, RSCEM recursively generates the spectral condition for the current band using the previously decoded bands, while PSCAM progressively constructs the spatial condition based on the pixels already decoded within the same band. These conditions are then used to obtain the probability distribution of the symbol at the current location and perform decoding. After the MSP of all bands has been decoded, the most significant bit plane is reconstructed and provided as a condition to the LSP branch; LSP is then decoded progressively using the same spectral order and diagonal spatial order. Finally, the decoded MSP and LSP are merged bitwise

$$X = X^M \cdot 2^d + X^L \tag{12}$$

to recover the output image.

IV. Experimental Results and Analysis

A. Experiment Setup

1) Experimental Data: To evaluate the performance of the proposed model, we conducted training and testing on the HyspecNet-11K [49], Chikusei [50], and Pavia dataset. HyspecNet-11K is currently the largest publicly available curated hyperspectral image dataset, derived from the EnMAP satellite. It contains 11 483 nonoverlapping image patches with a spatial size of 128×128 pixels and 224 spectral bands, and a ground sampling distance of 30 m. Following the author's official protocol, we adopted the "hard" mode to split the dataset, ensuring that the training and testing sets come from completely independent satellite orbits. During preprocessing, spectral bands severely affected by atmospheric water vapor were discarded, retaining 202 bands. To ensure data validity, pixel values were clipped to [0, 10 000], resulting in an effective dynamic range of approximately 14 bits.

The Chikusei dataset was acquired by the Headwall Hyperspec-VNIR-C imaging sensor and covers agricultural and urban areas in Chikusei City. It has a spatial size of 2517×2335, contains 128 spectral bands spanning 363–1018 nm, and is quantized at 16 bit. The Pavia dataset includes two subsets, Pavia University and Pavia Center, both acquired by the ROSIS sensor. The sensor provides 115 bands; after preprocessing, Pavia University contains 103 bands and Pavia Center contains 102 bands. Their spatial sizes are 610×340 and 1096×715, and the pixel values of both subsets range from 0 to 8000, corresponding to an effective quantization depth about 13-bit. Following LineRWKV, we cropped the data into 128×128 subimages. For Chikusei, we used a 9:1 train–test split. For Pavia, we used Pavia Center for training and Pavia University for testing.

2) Implementation Details: For the HyspecNet-11K dataset, parameter d for bit partition is set to 8. In the entropy model, both the PSCAM and RSCEM are configured with a feature dimension of 96. For plane condition aggregation, the number of hierarchical levels in the U-Net is set to 3. For the ParamNet, the mixture components are set to $N = 10$, and input feature dimensions for the MSP and LSP are set to 192 and 288, respectively, while the output feature dimension is set to 30. For our block strategy, we use a block size of $P = 32$. Finally, ANSs [48] are employed for entropy coding. Codes and pretrained weights are available at https://github.com/open-wit/LHIC-BP.

3) Evaluation Metric: We adopt bits per pixel per band (bpppb) as the primary evaluation metric. It is calculated as $B(X)/(S \times H \times W)$, where S denotes the number of spectral bands; H and W represent the height and width of each band, respectively; and $B(X)$ refers to the total bits of the compressed bitstream of the hyperspectral image X.

4) Training Details: All models are trained with the same configuration. During training, data augmentation is applied by flipping image patches horizontally and vertically with a probability of 0.5, followed by random cropping to 32×32 patches. Given the large volume of hyperspectral image, which leads to high memory usage and slow training, we adopt a two-stage training strategy. In the first stage, 16 consecutive

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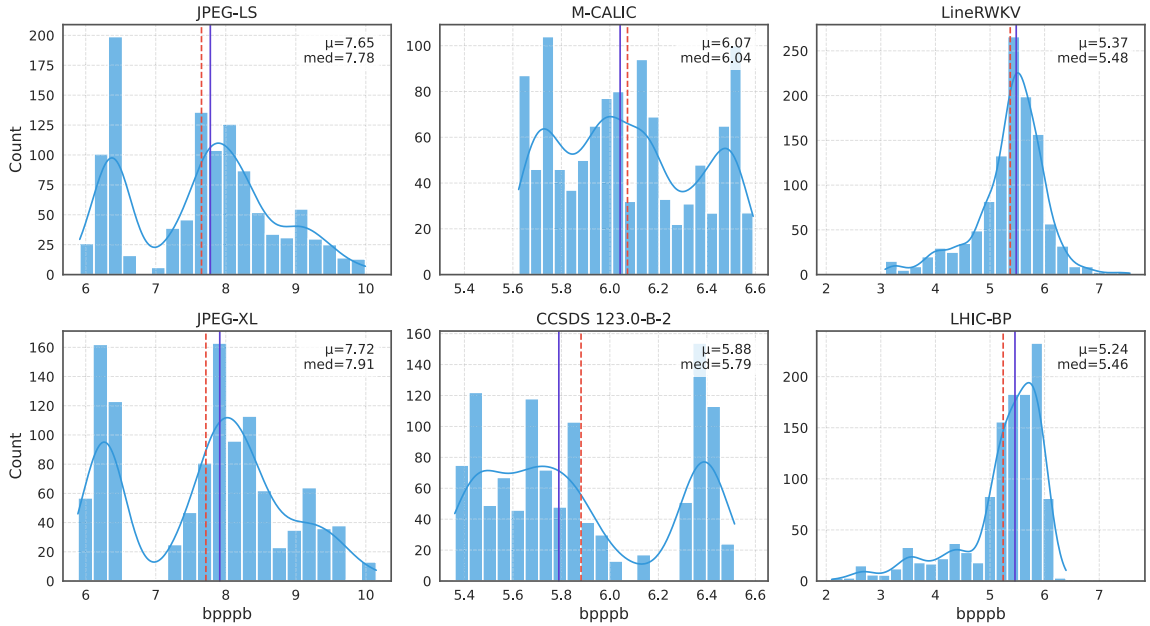


Fig. 9. Compression bitrate distributions (bpppb) across the test dataset for six lossless methods. Each subplot shows a blue histogram and blue KDE curve, with the mean marked by a red dashed line and the median by a green line. Reported values are the mean μ and the median med.

bands are randomly selected from the 202-band data, and the model is trained for 500 epochs with a batch size of 8. In the second stage, the model is fine-tuned on a longer spectral range of 96 with a batch size of 2 for 150 epochs. We use the AdamW optimizer [51] with an initial learning rate of $1e^{-4}$, linearly decayed to $1e^{-5}$, and fixed at $1e^{-5}$ during fine-tuning. All experiments are conducted with NVIDIA GeForce RTX 1080 Ti.

5) Baseline Codecs: We conduct a comprehensive comparison of the proposed framework with various image coders, including traditional natural image compression methods: JPEG-LS [52], JPEG 2000 [10], and JPEG XL [11], and learning-based methods: DLPR [46]. To address spectral redundancy in HSI, the original DLPR was modified by introducing a learnable linear coefficient for each band. For hyperspectral-specific methods, we choose traditional methods M-CALIC [53], CCSDS 123.0-B-2 [21], and RWA [20]. In addition, we compare our method with the predictive coding-based deep learning approach LineRWKV [23], where the reported results are taken from the original publication.

B. Main Results

1) Main Compression Results: In Table I, we report the main lossless compression results on the HyspecNet-11K dataset. LHIC-BP consistently achieves the best compression performance among all methods. In particular, it delivers a 9.71% bitrate saving over CCSDS 123.0-B-2 and a 2.45% bitrate saving over LineRWKV. These results indicate that neural network-based methods outperform handcrafted codecs and that conditional coding with end-to-end optimization is more effective than predictive coding for modeling complex spatial–spectral dependencies.

To highlight these advantages, Fig. 9 shows the bpppb distribution for six representative methods. Traditional codecs such as JPEG-LS and JPEG-XL yield higher bpppb, with wider and more variable distributions. Predictive coding methods such as M-CALIC and CCSDS 123.0-B-2 achieve lower bpppb and more concentrated distributions. LHIC-BP achieves the lowest average bpppb of 5.238 and remains robust across diverse samples. Compared with LineRWKV, it attains lower bpppb with a lower median and stronger concentration near the optimal region, demonstrating improved stability.

TABLE I

COMPRESSION RESULTS ON HYSPECNET-11K, THE MOST OPTIMAL RESULTS ARE INDICATED IN BOLD, WHILE THE SECOND OUTCOMES ARE UNDERLINED FOR CLARITY

Codec	bpppb ↓	Compression Ratio ↑
Tradition Image Codec		
JPEG-LS [52]	7.653	1.829
JPEG2000 [10]	7.892	1.774
JPEG-XL [11]	7.716	1.814
CCSDS123.0-B-2 [21]	5.881	2.381
M-CALIC [53]	6.073	2.305
RWA [20]	5.772	2.426
Learned Image Codec		
DLPR [46]	6.839	2.047
LineRWKV [23]	<u>5.370</u>	<u>2.607</u>
LHIC-BP* (Ours)	**5.238**	**2.673**

We further analyze the bitrate allocation between MSP and LSP to better understand the source of the compression gains. As shown in Table V, MSP contributes about 5% of the total bitrate, whereas LSP accounts for nearly 95%. This imbalance suggests that LSP dominates the coding cost and offers greater optimization potential, which motivates allocating more computational capacity to LSP.

2) Transfer to Other Sensor: Moreover, we investigated the generalizability of LHIC-BP. Specifically, the model trained

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TABLE II

COMPARISON OF COMPRESSION PERFORMANCE (BPPPB AND COMPRESSION RATIO) FOR MODEL TRAINED ON HYSPECNET-11K TRANSFER TO OTHER SENSORS; THE BEST RESULTS ARE IN BOLD, WHILE THE SECOND OUTCOMES ARE UNDERLINED FOR CLARITY

Codec	Chikusei		Pavia	
	bpppb ↓	Compression Ratio ↑	bpppb ↓	Compression Ratio ↑
Traditional Image Codec				
JPEG-LS	7.371	2.171	9.392	1.384
JPEG2000	7.772	2.059	9.850	1.320
JPEG-XL	7.490	2.136	9.538	1.363
M-CALIC	4.903	3.263	7.005	1.856
CCSDS123.0-B-2	4.661	3.433	6.697	1.941
Learned Image Codec				
LineRWKV(zero shot)	5.054	3.166	6.920	1.879
LineRWKV(finetune)	4.649	3.442	6.542	1.987
LHIC-BP* (zero shot)	4.786	3.343	6.891	1.887
LHIC-BP* (finetune)	**4.245**	**3.769**	**6.339**	**2.051**

TABLE III

AVERAGE CODING TIME PER BAND AND FLOPS PER SAMPLE OF OUR LHIC-BP. ENC. T. AND DEC. T. REPRESENT THE ENCODING AND DECODING TIMES, RESPECTIVELY

Codec	bpppb	Flops	Enc.T(s)	Dec.T(s)
CCSDS 123.0-B-2	5.881	–	–	–
LineRWKV	5.370	0.88M	0.0315	0.928
LHIC-BP* (Ours)	5.238	5.24M	0.0894	0.705

on HyspecNet-11K was fine-tuned to the Chikusei and Pavia datasets, and its performance on the corresponding test sets is reported in Table II. On both datasets, without fine-tuning, our method performs slightly worse than CCSDS but overall better than LineRWKV. With the same fine-tuning setting, LHIC-BP substantially outperforms both LineRWKV and CCSDS123.0-B-2 on Chikusei and Pavia, further confirming that our model achieves strong generalization and robustness when evaluated on data from different sensors.

3) Computational Complexity and Coding Time: We compared the computational complexity and the encoding and decoding time of LHIC-BP, LineRWKV, and CCSDS123.0-B-2, as summarized in Table III. Owing to the end-to-end conditional probability modeling, our method requires substantially higher FLOPs than LineRWKV, and its encoding time is approximately three times that of LineRWKV. It should be noted that, despite the higher encoding cost, our method achieves a clear improvement in compression performance and exhibits more stable results across multiple datasets. More importantly, LHIC-BP decodes noticeably faster than LineRWKV; on HyspecNet-11K, for example, decoding a single image can be nearly 1 min faster. This indicates a practical advantage. In compression, runtime is often one of the most important metrics. In general, data are encoded once and then distributed to users for repeated decoding, making decoding-side efficiency particularly critical. The decoding speedup of our method therefore has substantial practical value.

TABLE IV

ABLATION STUDY ON CONDITION INPUTS

Spatial	Spectral	Plane	MSP	LSP	Total
✓	✓	✓	0.239	4.999	5.238
✗	✓	✓	0.387	5.507	5.894
✓	✗	✓	0.793	5.889	6.282
✓	✓	✗	0.239	5.283	5.522

TABLE V

ABLATION STUDY ON ENTROPY MODEL

Spatial	Spectral	MSP	LSP	Total
PSCAM	RSCEM	0.239 (4.56%)	4.999 (95.44%)	5.238
PSCAM	Conv3d	0.271 (5.06%)	5.088 (94.94%)	5.359
Conv2d	RSCEM	0.254 (4.80%)	5.035 (95.20%)	5.289

C. Ablation Study

1) Impact of Condition Input: We first investigated the impact of conditional inputs on coding performance by evaluating spatial, spectral, and plane conditions. As shown in Table IV, results demonstrate that all three conditions improve compression efficiency, but to varying degrees: the spectral condition has the greatest influence, and removing it yields the worst performance (6.282 bpppb); the spatial condition ranks second (5.894 bpppb); and the plane condition has a minor effect. This hierarchy aligns with the characteristics of hyperspectral imagery. Dense spectral sampling introduces strong interband correlations and substantial spectral redundancy, whereas spatial redundancy is weaker due to the typically lower spatial resolution of hyperspectral sensors. The plane condition mainly affects LSP and has little impact on MSP.

To further examine these effects across the spectral dimension, we visualized the band-wise bpppb distribution for each variant in Fig. 10. The complete LHIC-BP consistently achieves the lowest bpppb values across all bands. Removing the spectral condition causes the most severe degradation, particularly in the higher bands; removing the spatial condition results in moderate degradation, most evident in the mid-to-high spectral range; and removing the plane condition leads to the smallest performance loss.

2) Impact of Entropy Model: In this section, we examined the influence of different architectural choices in the entropy model on coding performance. The proposed entropy model consists of two components: the PSCAM and the RSCEM. To evaluate their effectiveness, we replaced each component with alternative designs. PSCAM was substituted with a simplified structure consisting of a masked convolution followed by a 1×1 convolution (Conv2d), while RSCEM was replaced with a 3-D masked convolution (MaskConv3d) for spectral modeling. The results are summarized in Table V.

Replacing RSCEM with MaskConv3d increases the bpppb to 5.359, a 0.121 bpppb degradation. This decline can be attributed to the RWKV mechanism in RSCEM, which effectively captures long-range spectral dependencies that are

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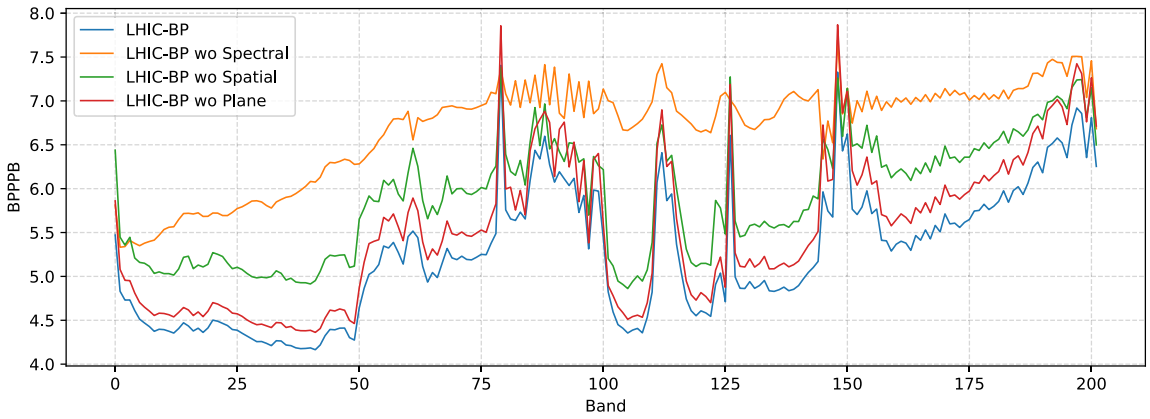


Fig. 10. Band-wise compression performance comparison in terms of bpppb on HyspecNet-11K.

TABLE VI
COMPARISON OF COMPRESSION PERFORMANCE (BPPPB AND COMPRESSION RATIO) FOR OTHER METHODS WITH BIT PARTITION

Codec	Year	MSP		LSP		Total	
		bpppb ↓	Compression Ratio ↑	bpppb ↓	Compression Ratio ↑	bpppb ↓	Compression Ratio ↑
JPEG-LS [52]	2000	0.989	6.067	7.426	1.077	8.415	1.664
JPEG2000 [10]	2001	1.163	5.159	8.068	0.992	9.231	1.517
JPEG-XL [11]	2019	0.744	8.065	7.878	1.015	8.622	1.624
CCSDS123.0-B-2 [21]	2019	0.571	10.508	7.076	1.131	7.647	1.831
LHIC-BP (Ours)	–	**0.239**	**25.105**	**4.999**	**1.600**	**5.238**	**2.673**

TABLE VII
COMPRESSION RESULTS ON HYSPECNET-11K WITH DIFFERENT PARAMETER d SETTINGS

d	MSP	LSP	Total	Enc.T(ms)
6	0.899	4.701	5.600	45.2
7	0.528	5.056	5.584	44.4
8	0.297	5.276	5.573	48.2
9	0.157	5.417	5.574	77.9
10	0.076	5.527	5.603	114.0

TABLE VIII
COMPRESSION RESULTS, AVERAGE GPU TIME PER BAND, AND MEMORY CONSUMPTION ON HYSPECNET-11K FOR ABLATION STUDY OF BIT PARTITION, CONDUCTED ON AN NVIDIA A800 (80 GB)

BP	bpppb	Enc.T(ms)	Memory(GB)
✓	5.573	20.1	2.06
✗	5.465	82.6	43.33

poorly modeled by the local operations of MaskConv3d. Similarly, substituting PSCAM with the simplified design results in a bpppb of 5.289, a 0.051 bpppb drop, indicating that the gated and attention-like mechanisms in PSCAM are essential for aggregating local spatial dependencies.

3) Impact of Bit Partition: In this section, we examine the impact of bit partition. First, we investigate the effect of the bit partition parameter d on compression performance and efficiency. The ablation study is conducted using a lightweight model with a feature dimension of 32. As shown in Table VII, experimental results show that $d = 8$ achieves the best tradeoff between bpppb and runtime. When d is too small, the increased randomness in the LSP makes accurate modeling difficult. As d increases, the number of symbols to be modeled in the autoregressive process grows, making probability distribution learning and fitting more challenging and imposing higher demands on model capacity. Moreover, a larger symbol range increases the computational cost of evaluating CDFs. Considering both modeling difficulty and cdf computation overhead, we select $d = 8$ as the optimal configuration for our proposed scheme on the HyspecNet-11k dataset.

Moreover, to assess the effectiveness of bit partition, we trained another network of comparable complexity without bit partition and achieved 5.465 bpppb. Although this result appears promising, the computational cost is unacceptable. Without bit partition, cdf computation during encoding requires GPU memory beyond the limit of a 1080Ti; thus, evaluation was conducted on an NVIDIA A800 (80 GB). As shown in Table VIII, with bit partition enabled, encoding requires 2.06 GB of memory and 20.1 ms per band, whereas disabling bit partition demands 43.33 GB of memory and 82.6 ms per band. In other words, bit partition provides about a 4× speedup while reducing memory consumption by roughly 20×. Given this comparison, the slight performance drop introduced by bit partition is entirely acceptable, demonstrating that it plays a crucial role in reducing complexity.

In addition, we integrate the bit partition strategy into various traditional handcrafted lossless compression methods and evaluate their compression rates on the dataset. As shown in Table VI, all of these conventional methods exhibit a large

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TABLE IX
MODEL PARAMETERS AND COMPUTATIONAL COMPLEXITY FOR EACH MODULE

	Total	MSP			LSP			
		ALL	PSCAM	RSCEM	ALL	PSCAM	RSCEM	U-Net
Total parameters	6.71M	1.98M	0.16M	1.66M	4.73M	0.32M	1.74M	2.54M
FLOPs	5.24M	1.51M	0.21M	1.31M	3.73M	0.32M	1.39M	2.02M

TABLE X
AVERAGE GPU TIME PER BAND AND BPPPB WITH DIFFERENT AUTOREGRESSIVE SETTINGS, OOM DENOTES OUT-OF-MEMORY DURING TRAINING, MEANING THAT THE GPU MEMORY REQUIREMENT EXCEEDS THE AVAILABLE CAPACITY, MAKING TRAINING INFEASIBLE

Block	Scan	Enc.T(s)	Dec.T(s)	bpppb
✓	Diagonal	0.048	0.458	5.573
✓	Raster	0.048	1.016	5.554
✗	Diagonal	0.109	1.873	OOM
✗	Raster	0.109	4.554	OOM

decline in performance. This finding highlights that specialized design is required when applying bit partition to maintain compression efficiency.

4) Impact of BDAS: Here, we compare the impact of different autoregressive strategies on coding time. Specifically, for the proposed BDAS, we conduct experiments with the same setting in Section IV-C3 to evaluate the block-wise strategy and diagonal autoregression separately. As shown in Table X, scanning order does not affect encoding time because our framework employs parallel one-step encoding for each band. In contrast, decoding is inherently sequential; thus, diagonal scanning reduces the number of decoding steps and accelerates decoding, at the cost of a slight compression-performance degradation that is considered acceptable. Moreover, the block-wise strategy substantially accelerates both encoding and decoding. Beyond runtime, the high spatial–spectral complexity of hyperspectral imagery also causes substantial GPU memory consumption during training. For example, when training with a batch size of 8 on LSP inputs of size $(16, 128, 128)$, replacing full-resolution inputs with 32×32 blocks reduces peak memory usage from 240 to 15 GB (93.75% reduction). This improvement enables practical model training on custom datasets even with low-cost GPUs, thereby enhancing the accessibility of hyperspectral compression research.

D. Complexity Analysis of Each Module

We conduct a comprehensive complexity analysis of each module from two perspectives: the total number of parameters and floating-point operations (FLOPs). As shown in Table IX, most parameters and FLOPs are allocated to the LSP module (4.73 M parameters and 3.73 M FLOPs), whereas the MSP module remains relatively lightweight (1.98 M parameters and 1.51 M FLOPs). This design is consistent with the characteristics of the subimages. MSP contains the major intensity information and is easier to model, whereas LSP carries fine-grained residuals and noise-like components that require a stronger network, such as the U-Net for plane condition extraction, for effective context modeling. Notably, although the LSP branch increases complexity, it brings significant compression gains, indicating that LHIC-BP achieves an effective tradeoff between complexity and performance.

E. Limitations

Although our method explores the limit performance of neural networks in lossless compression of HSI and adopts various approaches to reduce coding complexity, the bit partition design introduces relatively complex prior modeling of bit planes. Modeling each band separately requires considerable GPU memory. Moreover, due to the autoregressive nature, decoding band by band results in significant time consumption for images with numerous bands.

V. CONCLUSION

In this article, we propose LHIC-BP, the first end-to-end optimized framework for lossless compression of HSI. Our method employs conditional coding to fully utilize the optimization capabilities of neural networks while incorporating a bit partition strategy to improve algorithmic feasibility. To address the unique characteristics of hyperspectral image, we propose PSCAM and RSCEM that together form an entropy model capable of effectively removing both spatial and spectral redundancies. Experimental results demonstrate that our method surpasses all existing approaches in terms of compression performance. For future work, we will further investigate the performance limits of lossless hyperspectral compression while continuing to optimize the algorithm to reduce complexity and achieve practical compression.

ACKNOWLEDGMENT

The numerical calculations in this article have been done on the supercomputing system in the Supercomputing Center of Wuhan University.

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